Data and Code Availability

Data

OBELiX (Therrien et al., *Digital Discovery* 2026, 5, 910–918; DOI 10.1039/D5DD00441A) is licensed CC BY 4.0 and may be redistributed with attribution. Its official train/validation/test split was reused exactly; the 449-row train+validation union is this study's development population. The 113-row official test split was not used for fitting, tuning, internal validation, external-transfer evaluation or any reported predictive-performance estimate; one historical label-compatibility sensitivity read its labels and is disclosed in the Supplementary Information, and the reported compatibility analysis uses development rows only.

The Hargreaves et al. lithium solid-electrolyte conductivity database (Hargreaves et al., *npj Comput. Mater.* 2023, 9, 9; DOI 10.1038/s41524-022-00951-z) is not freely redistributable: research and academic use with citation of the University of Liverpool is permitted, and commercial re-use requires a separate licence from the University. Because these terms do not grant redistribution, no file from which the external conductivity values can be read is included in the public release: the raw database file, the row-level overlap ledger and the formula-level strict (204) and relaxed (239) cohort files are not included in the public release, and the per-formula external targets and residuals are removed from the archived external-transfer prediction records (every prediction and every aggregate metric is kept). The release records the source URL, retrieval date (2026-09-21), file size, row count and SHA-256 hash of the file used, the SHA-256 hash of every withheld or redacted file, and a script that rebuilds the cohorts and ledger from a lawfully obtained copy, restores the redacted fields and verifies every result against the recorded hashes (data/RECONSTRUCTION.md, scripts/release/restricted_data.py).

The OBELiX source files and processed split, model predictions, evaluation records and all sensitivity-analysis outputs are included in the release. No new experimental measurements or atomistic-simulation data were generated.

Code

The code for cohort construction, nested internal validation, one-shot external transfer, every post hoc sensitivity analysis, and figure, table and manuscript generation is maintained in a version-controlled repository and is released under the MIT licence. The reproducibility archive contains the source code, the hash-locked protocol and analysis specifications, the authoritative analysis records, a pinned Python environment, a SHA-256 manifest of every file, and scripts that rerun the analyses and compare the results with the recorded values.

Public deposit. The public repository URL and archival DOI have not yet been created; they will be inserted here before submission. Until the deposit is complete, the code and reproducibility archive are available from the corresponding author on reasonable request.